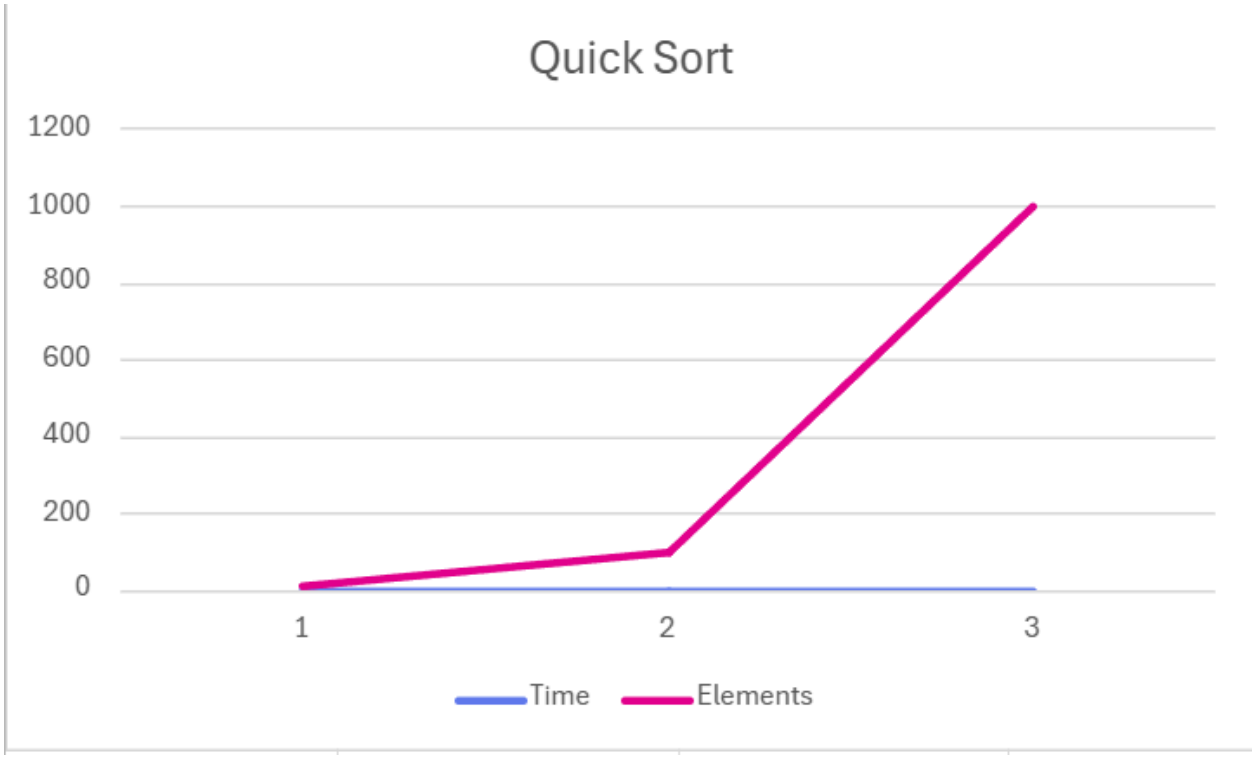
- **Introduction** – Linked lists are arrays that point to other elements in the list. Linked lists point to the next element in a list, while doubly linked lists point to the previous and next element in a list.

- **Programmer's Guide –** For this assignment I used an array, and sorted through it with a Quicksort algorithm. This algorithm treated the array as a linked list as it checked its elements in accordance with each other. If the element after the current one was smaller than it, the Quicksort algorithm swapped their positions. The Quicksort function takes an array argument and two nodes to compare to each other. It then uses the partition variable to split the array into two halves to sort. This Quicksort function can be used to sort through large datasets for example, putting files in order by date. The Heap method takes an array as an argument and two nodes to compare. It determines which node is larger until it's at the top, removes it, and repeats until all the nodes have been sorted in accordance to each other. This Heapsort function can also be used to sort through large databases, and can be used to find specific elements like the 1st and last element. The print method displays the sorted arrays to the console if it has less than 100 elements. This is useful for keeping the user updated and debugging clear, while also not cluttering up their screen. The time and sort operation times how long it takes an algorithm to sort the data and prints it to the screen. It takes an array, the size of the array, and an output message as arguments. This is useful for tracking the time complexity of an algorithm.

- **Analysis –** O(n^2) is the worst-case time complexity for Quicksort but for my program it hit O(n log n). O(n log n) is the worst-case time complexity for Heapsort as well, and in my program it didn't exceed this.






- Does your calculated run-time complexity match the measured run-time complexity?  Yes and no. Quicksort performed better than expected while Heapsort stayed within expectations.
    - Under which dataset sizes does each algorithm perform best? Why or why not?  Does any algorithm perform best under all cases? They both seem to perform similarly under both dataset sizes, however, Heapsort seems to perform well in all cases.